

ModelNet40 Point-Cloud Upsampling — Full Thesis Report

Equal-N Geometry + PointNet++ SSG Retrain Classification

- Companion PPTX has 28 slides: ModelNet40_PointNet2_Full_Report.pptx
- Classifier: pointnet2_cls_ssg from scratch per variant · 200 epochs · Adam 0.001 · seed 42
- NOT finetune of Original-1024 checkpoint; NOT training the upsampler in classification stage
- Generated: 2026-08-04 11:52:38 UTC

Why Retrain (English)

- Retrain = train PointNet++ SSG from scratch on each cloud variant with matched num_point
- Isolates representation (density/distribution) under a fair fixed backbone
- Matching num_point avoids input-size mismatch; from-scratch avoids transfer from another density
- PointNet++ SSG is the project standard ModelNet40 backbone
- Upsamplers only generate clouds; retrain refers to the classifier only
- Fair protocol: same arch/hparams, independent checkpoints; stricter frozen-classifier protocol not done

■■■■■■■■■■ (■■)

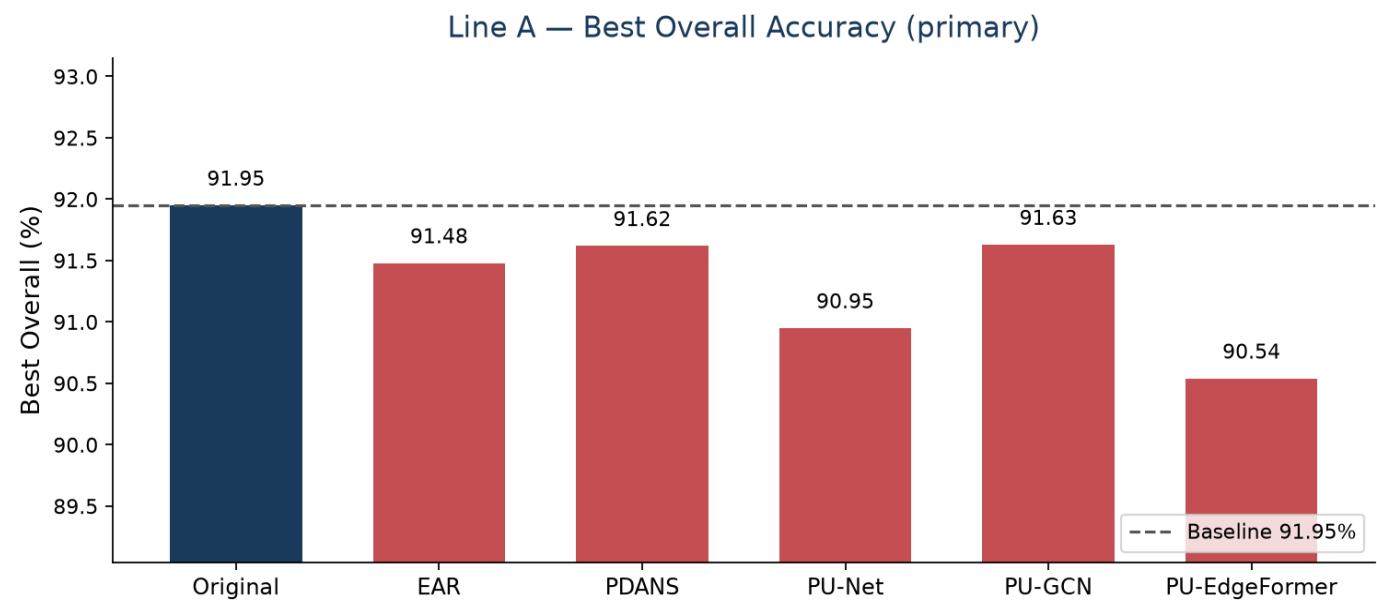
- ■■■■■■■■■■■■ PointNet++ SSG■■■■ num_point■■■■ finetune Original-1024
- ■■■■■■■■■■——■■■■■■■■■■
- ■■■■■■ backbone ■■■■■■■■■■■■■■■■
- ■■■■■■ test instance accuracy ■ best_model■■■■■■■■■

Baselines — Best / Final OA

Baseline	Pts	Best OA	Final OA
Original 1024	1024	91.95%	91.31%
Downsampled ×4	256	90.85%	90.46%
Mesh-ref 256	256	90.88%	90.38%
Mesh-ref 4096	4096	92.00%	91.18%

Line A vs Mesh-ref 4096

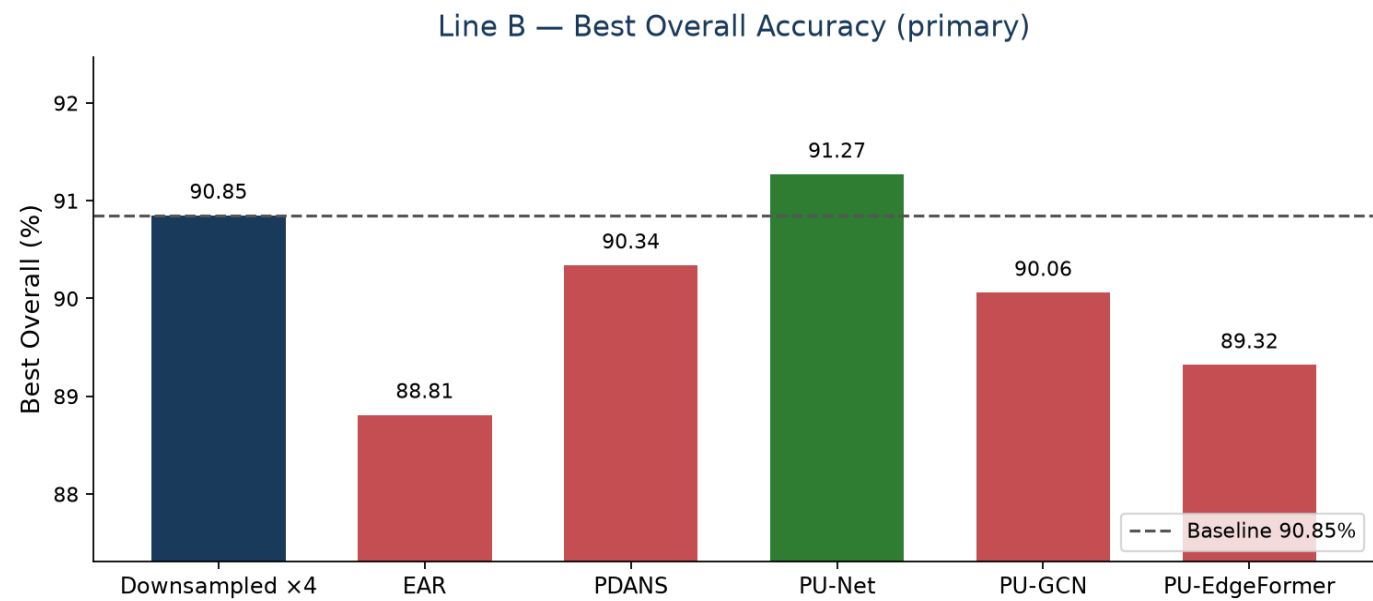
Method	Best OA	Δ vs Mesh-ref 4096	Final OA
Mesh-ref 4096	92.00%	+0.00 pp	91.18%
PU-GCN	91.63%	-0.36 pp	91.00%
PDANS	91.62%	-0.38 pp	90.87%
EAR	91.48%	-0.52 pp	90.85%
PU-Net	90.95%	-1.05 pp	90.55%
PU-EdgeFormer	90.54%	-1.46 pp	90.13%



Line B Recovery

Method	Pts	Best OA	Δ vs Down	Gap vs Orig
Downsampled $\times 4$	256	90.85%	+0.00 pp	-1.10 pp
EAR	1024	88.81%	-2.04 pp	-3.14 pp
PDANS	1024	90.34%	-0.51 pp	-1.61 pp
PU-Net	1024	91.27%	+0.42 pp	-0.68 pp
PU-GCN	1024	90.06%	-0.79 pp	-1.89 pp
PU-EdgeFormer	1024	89.32%	-1.53 pp	-2.63 pp

- PU-Net 91.27% vs Down 90.85% (+0.42 pp); still -0.68 pp below Original 91.95%



Equal-N Geometry Summary

Line	Method	Pts	Ref	CD	HD	NUC	Δ NUC
A	Mesh-ref 4096	4096	Mesh-ref 4096	0.0000	0.0000	0.6419	0.0000
A	EAR	4096	Mesh-ref 4096	0.0547	0.1154	0.8156	0.1738
A	PDANS	4096	Mesh-ref 4096	0.0468	0.1135	0.6612	0.0193
A	PU-Net	4096	Mesh-ref 4096	0.0497	0.1112	0.6052	-0.0366
A	PU-GCN	4096	Mesh-ref 4096	0.0423	0.0932	0.5204	-0.1215
A	PU-EdgeFormer	4096	Mesh-ref 4096	0.0467	0.1109	0.6754	0.0336
B	Mesh-ref 256	256	Mesh-ref 256	0.0000	0.0000	2.1265	0.0000
B	Downsampled $\times 4$	256	Mesh-ref 256	0.1333	0.2049	2.0933	-0.0332
B	Original	1024	Original 1024	0.0000	0.0000	1.0836	0.0000
B	EAR	1024	Original 1024	0.0767	0.1932	0.9653	-0.1183
B	PDANS	1024	Original 1024	0.0627	0.1905	1.0528	-0.0307
B	PU-Net	1024	Original 1024	0.0646	0.1083	1.0801	-0.0035
B	PU-GCN	1024	Original 1024	0.0558	0.1514	1.5240	0.4404
B	PU-EdgeFormer	1024	Original 1024	0.0619	0.1776	1.1225	0.0390

Findings & Conclusions

- No Line A upsampler beats Mesh-ref 4096 (92.00%) or Original (91.95%)
- Mesh-ref $\approx$ Down/Original (+0.03 / +0.05 pp)
- Line B PU-Net partial recovery only (+0.42 vs Down; -0.68 vs Orig)
- Better geometry $\neq$ automatic OA gains
- Limitations: best on test (no val); independent retrain $\neq$ frozen single classifier

Line A qualitative — airplane

Line A · airplane · airplane_0627

Same sample · Same camera · Same axis limits · Same point-size rule | Color = depth (not error)

Original
1024 pts

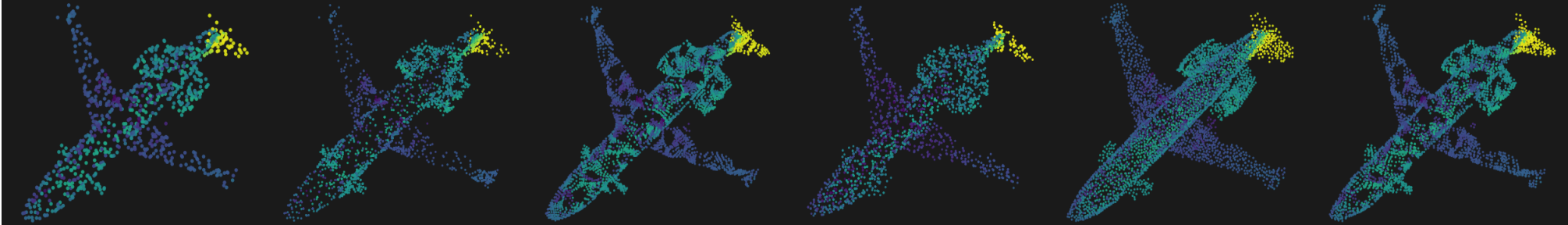
EAR
4096 pts

PDANS
4096 pts

PU-Net
4096 pts

PU-GCN
4096 pts

PU-EdgeFormer
4096 pts



Line B qualitative — airplane

